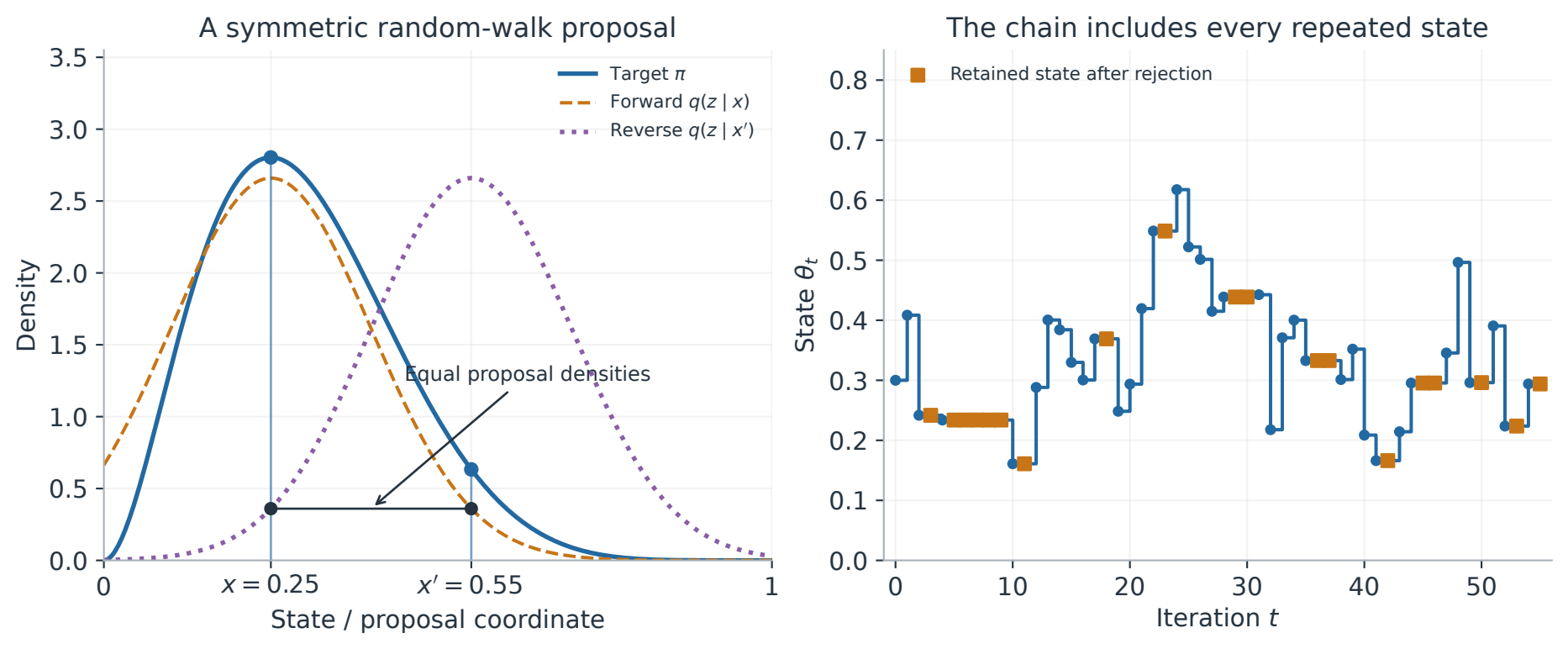
A symmetric random-walk proposal
Target π
Forward $q(z \mid x)$
Reverse $q(z \mid x')$
Equal proposal densities
Density
State / proposal coordinate
$x = 0.25$
$x' = 0.55$
The chain includes every repeated state
Retained state after rejection
State θ_t
Iteration t